

SRI VENKATESHA MANI NARAYANAN

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Professional Summary

Software Engineer with 2 years of production experience across Full Stack Development, Cloud Native architectures and AI/ML applications. Proficient in Python, JavaScript, TypeScript, and Java with experience building production applications using FastAPI, Flask, Next.js, React.js, Node.js, and Express.js. Hands-on experience with Docker, Kubernetes, GCP, and AWS including EC2, Lambda, ECS, Fargate, S3, and SQS. Specializing in LLMs, RAG, AI agents, LangChain/LangGraph, MCP, and Vector Database.

Technical Skills

Languages: Python, JavaScript, TypeScript, Ruby, Java, SQL, HTML, CSS.

Frameworks & API: FastAPI, Flask, Node.js, Express.js, Next.js, React.js, Rails, REST APIs, GraphQL, WebSockets

AI / ML: AWS Bedrock, LangChain, LangGraph, Model Context Protocol (MCP), RAG, Vector Search, Agentic Workflows

Database & Messaging: PostgreSQL/pgvector, MySQL, MongoDB, Redis, Kafka, Snowflake.

Cloud & Infra: AWS(EC2, Lambda, ECS, Fargate, S3, SQS, RDS, EKS), Docker, Kubernetes, Terraform, Ansible

DevOps: CI/CD, GitHub Actions, Jenkins, CloudWatch, Prometheus, Grafana

Work Experience

Software Engineer | [Rattlesnake Ramble Organization](#) | Boulder, CO

Aug 2025 – Aug 2026

- Led a team migrating a **legacy PHP** application to a modern **PERN platform**, rebuilding **20+ features** and executing an **ETL migration to PostgreSQL** while architecting serverless workloads with **AWS Lambda and SQS**, driving **78% registration growth**.
- Resolved a critical **payment-state failure** by implementing **async PayPal webhooks with Amazon SQS and Lambda**, adding server-side verification, **idempotency**, **retries**, and **reconciliation** to maintain accurate payment state across **70+ registrations**.
- Migrated **2,500+ historical racer records spanning 20 years** from legacy **MySQL to PostgreSQL** through an **ETL pipeline**, transforming schemas, preserving **relational integrity**, and performing post-migration validation for production cutover.
- Deployed a serverless production architecture using **AWS Lambda with Amazon S3** for persistent media storage, implementing a **CI/CD pipeline with GitHub Actions** to automate builds and deployments and reduce production deployment time by **75%**.
- Developed an **LLM-powered admin assistant integrated into the React dashboard**, using **tool calling** to answer natural-language queries across registration and payment data, consolidating **7+ administrative workflows** into one conversational interface.

Software Engineer – AI & Full Stack | [Biocollate](#) | Boulder, CO

Aug 2024 – May 2025

- Owned end-to-end development of an **AI-powered LIMS/ELN**, containerizing **React and FastAPI** services and deploying them on **AWS ECS/Fargate** while integrating **RAG and agentic workflows** across **27 SBOL/LabOP APIs** serving **30+ lab researchers**.
- Built dependency-aware lifecycle management for **100+ biological components**, implementing **recursive traversal** to preserve component hierarchies, version history, and **RBAC**, reducing component lookup and **dependency-tracing time by 60%**.
- Engineered **LabOP-based ELN workflows** for shared protocol authoring and execution, implementing **versioned records**, **audit trails**, and **dual-signature verification** with independent witnesses, **reducing experiment documentation time by 50%**.
- Implemented **CatalystX**, a **RAG-powered AI assistant** using **LangChain**, LLM APIs, embeddings, and **PostgreSQL/pgvector** to semantically retrieve components and experiments, eliminating manual searches for **90% of routine researcher queries**.
- Developed **agentic workflows using LangGraph and LLM tool calling** across Benchtop Lab Assistant services, enabling **CatalystX** to trace component dependencies and affected experiments in **one natural-language request versus 5+ manual lookups**.

Software Development Intern | [ITJobxs](#) | Chennai, India

May 2024 – Aug 2024

- Delivered an end-to-end interview-experiences platform with **React and Express.js**, engineering **authentication**, **user verification**, and **MySQL-backed REST APIs** that enabled **500+ users** to securely publish and search interview experiences.
- Prevented **20–50 fake registrations per day** by implementing **O(1) hash-based email domain validation** and **Google reCAPTCHA** checks, intercepting bot traffic before account creation and MySQL insertion while strengthening signup security.
- Deployed the production application on **AWS EC2** with **Route 53** DNS routing and **CloudFront CDN** caching, configuring **Nginx as a reverse proxy** for the Node.js/Express backend and reducing asset load times by **35%**.

Projects

RecipeRealm – Recipe Sharing Platform

- Built RecipeRealm, a full-stack recipe-sharing platform with JWT-secured user accounts and social interactions, Dockerizing MERN services and deploying on AWS EC2 with S3-backed image and video storage while achieving 85% Jest test coverage.

AI-Generated Text Detection

- Fine-tuned BERT, DistilBERT, and GPT-2 using PyTorch and Hugging Face to detect AI-generated text across complex, multi-source datasets, optimizing hyperparameters and engineering a model pipeline that achieved 82.7% evaluation accuracy.

Education

MS in Computer Science | [University of Colorado Boulder](#) | 3.91/4.0 | Boulder, CO

2023 – 2025

B.Tech in Computer Science & Engineering | [SRM Institute of Science & Technology](#) | 9.29/10.0 | Chennai, India

2019 – 2023